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Impact of Luminescent Coupling on Perovskite-Silicon Tandem External Quantum Efficiency Quantified by Comprehensive Opto-Electronic Simulation

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ABSTRACT

In this work, the effects of luminescent coupling (LC) on the external quantum efficiency (EQE) of perovskite-silicon tandem (PST) solar cells are quantified by means of monochromatic transient photocurrent measurements and comprehensive optoelectronic simulations that take into account both optical and electrical coupling of the subcells. It is shown that, at short wavelengths, a similar response results from both LC and silicon bottom-cell shunts. The two contributions can be discriminated and quantified based on bias voltage and light intensity-dependent measurements. Such measurements were conducted on state-of-the-art PST cells and agree well with the behavior predicted by the simulations. For the case of polychromatic EQE simulations, a quenching of the LC effects with decreasing concentration of mobile ions is found, which is explained in terms of ion-modulated recombination via bulk defects.

1 | Introduction

With single-junction cells approaching their practical efficiency limits, research is increasingly focusing on multijunction approaches [1, 2]. A promising technology is the perovskite-silicon tandem (PST) solar cell, now reaching power conversion efficiencies of >35% [3]. As the perovskite top cell operates ever closer to the radiative limit, luminescent coupling (LC) effects (bottom cell reabsorption of photons that are emitted in the top cell; cf. Figure 1a) become increasingly important to consider, such as the beneficial impact on current matching in bottom-limited tandems [4–6]. Although these effects are generally beneficial in the context of solar cell operation, the additional energy-transfer pathway by optical means can, however, complicate the interpretation of certain characterization experiments. One important measurement, which is highly affected by LC, is the evaluation of subcell external quantum efficiencies (EQE), as LC breaks the assumption of pure electronic coupling through the recombination junction [7]. While this problem has attracted a great deal of interest in the III–V multijunction solar cell community [8–14], it is less explored in perovskite-based tandems. Furthermore, PSTs pose additional challenges when determining the subcell EQE, such as the presence of mobile ions in the perovskite [15].

The effects of LC are conventionally expressed via a LC efficiency η_{LC} that relates the additional photocurrent in the bottom cell due to LC to the radiative recombination current in the top cell [16], and which is usually determined as a fit parameter in equivalent circuit (EC) modeling of subcell characteristics [9]. More recently, also experiments with photoreflectance [17], three-terminal devices [18], modulated photocurrent measurements [19], or a combination of intensity-dependent photoluminescence (PL) and photocurrent measurements with absolute photoluminescence

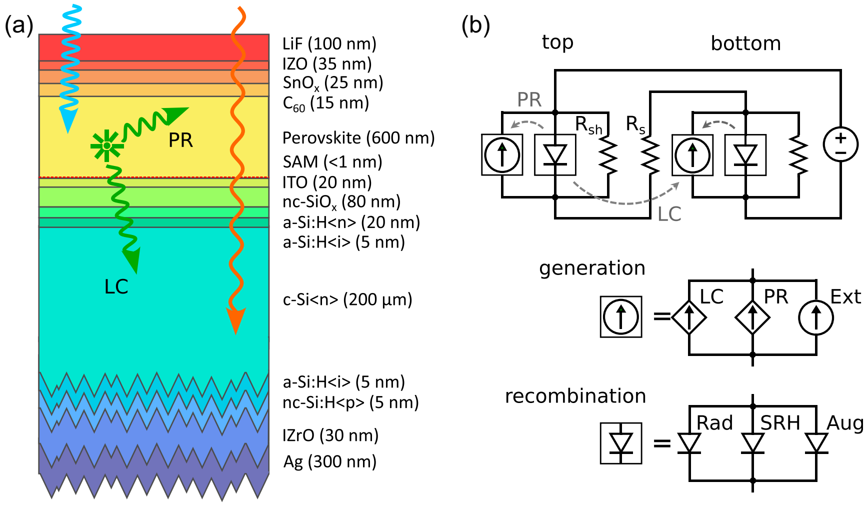

FIGURE 1 | (a) Schematic representation of the backside-textured PST solar cell device stack layout and of the reabsorption processes for photons emitted by the perovskite absorber. (b) The EC describing the idealized electronic transport model. The lumped current source consists of three current sources for generation due to external sources, PR, and LC. The lumped recombination diode includes radiative, Shockley–Read–Hall (SRH), and Auger recombination. LC and PR are dependent current sources, interacting with the radiative recombination diode as depicted by the dashed arrows.

quantum yield (PLQY) [7] have been used to assess this quantity. The main contribution to the LC efficiency, the optical transfer from top to bottom cell, has been approximated using optical models ranging from ray-optical estimates in thick absorbers [20] to transfer-matrix-method (TMM) approaches in thin films [21] or a combination of ray-tracing with TMM [6].

Here, the quantification and analysis of LC effects in PST solar cells are addressed using a comprehensive opto-electronic simulation framework. At the core of the approach lies a consistent and rigorous optical model for absorption and luminescence in solar cell devices under consideration of coherent and incoherent photon propagation including scattering at interface textures [22]. The results are fed to electrical models ranging from simplified ECs [23] to a full-fledged drift-diffusion charge transport framework including mobile ion physics at steady state [24] and in transient mode [25, 26]. The optical model provides local absorption, emission, and reabsorption rates throughout the device. These rates act as source/sink terms in the electronic model, allowing for a consistent treatment of both photon recycling (PR) and LC contributions under arbitrary optical and electrical bias. Fitting this model to measured EQE data from state-of-the-art PST solar cells fabricated at EPFL PV-Lab [27, 28] as a function of illumination intensity or bias voltage then allows us to identify and separate the various mechanisms responsible for the observed behavior. Most importantly, the model enables discrimination between contributions to the short-wavelength response that are due to either LC or low shunt resistance in the bottom cell [8, 12, 29].

The article is organized as follows. In Section 2, details on the multiscale simulation approach are provided. The intensity- and bias-dependent EQE measurement procedure is reported in Section 3, and the results of both are compared and analyzed in Section 4.

2 | Simulation Approach

The multiscale optical model used consists of two coupled components: a recently developed dyadic Green tensor formalism [30] describes emission, reabsorption, and coherent propagation within the thin films. It is fully compatible with local and global detailed balance and is free from unphysical divergence in the dipole power emitted within the absorbing media. This picture is coupled to a net-radiation type optical model for emission, reabsorption, and incoherent propagation in optically thick media, such as the Si absorber, where additionally incoherent light scattering can be taken into account [22]. In addition to the photogeneration rate $\mathcal{G}_{ext}$ due to external irradiation with photon flux ϕ_{ext}, numerical evaluation of the coupled models provides the normalized profile of local internal emission $\mathcal{R}_{em,0}$ and reabsorption $\mathcal{G}_{reabs,0}$ as well as the local photon flux $\mathcal{S}_0$ due to internal emission, which needs to be scaled using the local quasi-Fermi level splitting (QFLS) $\Delta\mathcal{F}$. The latter is provided by the electronic transport model and constitutes the electronic to optical coupling. Explicit expressions of the internal rates as a function of the Green tensor can be found in Ref. [23]. The profiles of local emission and reabsorption rates, as well as the photon energy flux associated with propagating modes originating from internal emission in the stack displayed in Figure 1a, are given in Figure 2a,b, respectively. The local rates reveal not only signs of coherent emission in the perovskite, but also significant reabsorption in the top cell, while the Poynting vector shows a strong asymmetry favouring coupling of light to the silicon bottom cell over out-coupling at the top. As shown in Figure 2c, PR dominates over LC in this case. LC amounts to 36% of the photons emitted by the top cell, which is very close to the LC efficiency of 35% found experimentally in [7] for a very similar structure with large ratio of bottom to top emission. However, in [7], the LC efficiency includes also nonradiative losses via the radiative efficiency as a factor in PLQY.

To go beyond optics and capture the impact of electronic losses on the extraction of photogenerated charge carriers, two approaches of different rigor are considered. The most general case is covered by a drift-diffusion-Poisson model to properly take into account any nonidealities in terms of electronic transport, as well as mobile ions, in steady-state, transient, and small-signal AC regimes [24] as implemented in the device simulation tool SETFOS [31]. In this

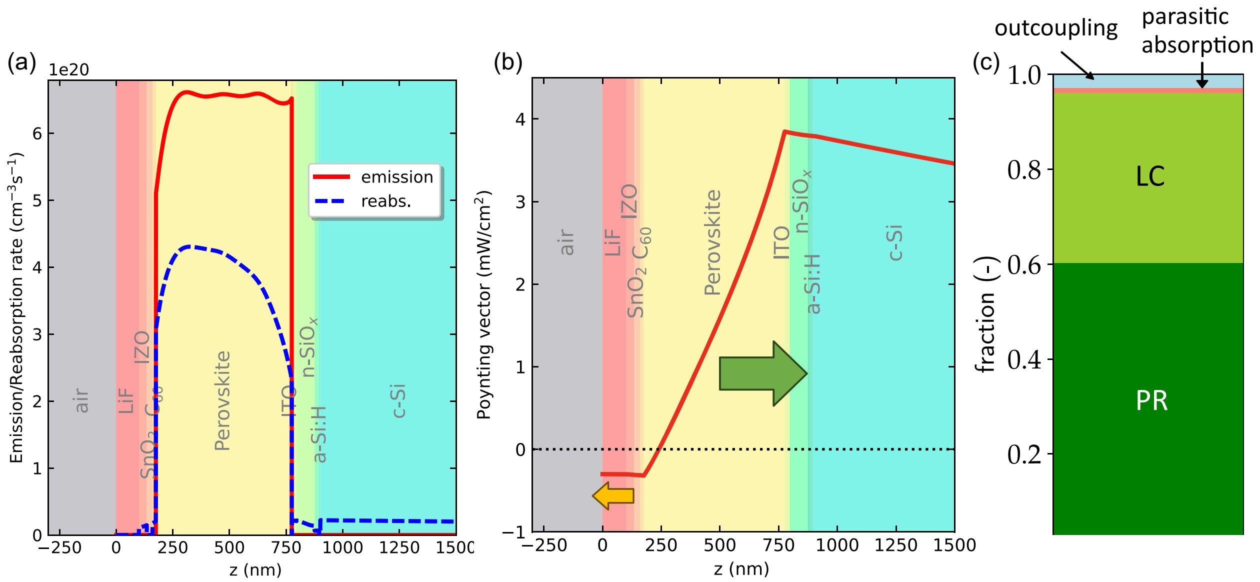

FIGURE 2 | Optical assessment of quantities related to internal emission: (a) Profiles of emission and reabsorption rates (to be scaled by the Boltzmann factor with the QFLS). (b) Poynting vector of propagating modes from internal emission, showing a dominant coupling of light to the silicon bottom cell. (c) In this device, PR dominates over LC, while the shares of outcoupling and parasitic absorption are marginal.

context, the optical to electronic coupling enters the equations through radiative recombination (emission) and secondary photogeneration (reabsorption) rates in the continuity equations for electrons and holes. Similar combinations of optical and charge transport models have been used previously to address LC in III–V multijunction solar cells [32–34], albeit with less rigor regarding the consideration of the optical density of states in the emission and the impact of longitudinal modes, and without light scattering. On the other hand, the advanced optical model can also be used to parametrize the LC term in a simplified EC approach which is suitable for extensive parameter studies to obtain qualitative insight and for the fitting of experimental data. The generation currents in the EC scheme shown in Figure 1b include the photocurrents due to external illumination determined based on the active layer absorptance as well as the nonlinear dependent current sources for PR and LC, where the optical to electronic coupling enters in the definition of the corresponding saturation current prefactors obtained from the spatial integration of $\mathcal{G}_{\mathrm{reabs},0}$ with appropriate choice of source domain

$$J^i_{\mathrm{gen}} = J^i_{\mathrm{ext}} + J^i_{\mathrm{PR}} + J^i_{\mathrm{LC}}, \qquad i = \mathrm{top, bot} \tag{1}$$

$$J^i_{\mathrm{ext}} = \eta_{\mathrm{col}} \int \mathrm{d}E_\gamma A^i(E_\gamma)\phi_{\mathrm{ext}}(E_\gamma) \tag{2}$$

$$J^i_{\mathrm{PR}} = J^i_{\mathrm{0PR}}\left[\mathcal{G}^{i\to i}_{\mathrm{reabs},0}\right]\left\{\exp\left(\frac{qV^i}{k_\mathrm{B}T}\right) - 1\right\} \tag{3}$$

$$J^i_{\mathrm{LC}} = J^i_{\mathrm{0LC}}\left[\mathcal{G}^{j\to i}_{\mathrm{reabs},0}\right]\left\{\exp\left(\frac{qV^j}{k_\mathrm{B}T}\right) - 1\right\} \tag{4}$$

where η_{col} is the carrier collection efficiency and $V^i, i \in \{\mathrm{top, bot}\}$ are the internal subcell voltages. The recombination term contains the contributions due to radiative, SRH and Auger recombination, which are expressed in the usual way via the respective dark saturation currents and ideality factors

$$J^i_{\mathrm{rec}} = J^i_{\mathrm{rad}} + J^i_{\mathrm{SRH}} + J^i_{\mathrm{Aug}}, \qquad i = \mathrm{top, bot} \tag{5}$$

$$J^i_{\mathrm{rad}} = J^i_{\mathrm{0rad}}\left[\mathcal{R}^i_{\mathrm{em}}\right]\left\{\exp\left(\frac{qV^i}{k_\mathrm{B}T}\right) - 1\right\} \tag{6}$$

$$J^i_{\mathrm{SRH}} = J^i_{\mathrm{0SRH}}\left\{\exp\left(\frac{qV^i}{2k_\mathrm{B}T}\right) - 1\right\} \tag{7}$$

$$J^i_{\mathrm{Aug}} = J^i_{\mathrm{0Aug}}\left\{\exp\left(\frac{3qV^i}{2k_\mathrm{B}T}\right) - 1\right\} \tag{8}$$

As compared to the framework presented in Ref. [35], we neglect the current component related to reverse bias breakdown and the subcell voltage-independent LC current related to PL, as both are minor relative contributions at the operating point of the tandem solar cell considered here. Also, the PR and LC saturation currents may contain respective collection efficiencies to enable bridging the optical limit—where transport is idealized and quasi-Fermi levels are flat—and compact models that consider nonradiative losses also at short circuit. With respect to the above optical currents, the LC efficiency is defined as[1]

$$\eta^{j\to i}_{\mathrm{LC}} = \frac{J^i_{\mathrm{0LC}}\left[\mathcal{G}^{j\to i}_{\mathrm{reabs},0}\right]}{J^j_{\mathrm{0rad}}\left[\mathcal{R}^j_{\mathrm{em},0}\right]} \tag{9}$$

Due to the small overlap of the bottom cell emission spectrum with the top cell absorptance, only $\eta^{\mathrm{top}\to\mathrm{bot}}_{\mathrm{LC}}$ is evaluated. The total current density per subcell then reads

$$J^i = -J^i_{\mathrm{gen}} + J^i_{\mathrm{rec}} + J^i_{\mathrm{sh}}, \qquad i = \mathrm{top, bot} \tag{10}$$

where shunt conductance is considered as an additional subcell current element via the internal parallel resistance R^i_{sh}, $i \in \{\mathrm{top, bot}\}$ in the form of the shunt currents $J^i_{\mathrm{sh}} = V^i/R^i_{\mathrm{sh}}$. The subcell voltages are related to the externally applied bias voltage V and the global series resistance R_s via

$$V = V^{\mathrm{top}} + V^{\mathrm{bot}} + R_\mathrm{s} \cdot J \tag{11}$$

$$J = J^{\mathrm{top}} = J^{\mathrm{bot}} \tag{12}$$

Equations (11) and (12) form a nonlinear system of equations for the subcell voltages as a function of the external illumination ϕ_{ext} and the applied bias voltage V, which is solved using a Brentq algorithm, similar to the optical-limit computations in Refs. [36] and [23]. The model described above is computationally highly efficient, as the expensive determination of the nonlocal reabsorption rate needs to be performed only once for the parametrization of the current prefactors related to radiative processes.

3 | Measurement Procedure

Irradiance-dependent monochromatic EQE was measured on five different PST samples with nominally identical structure as shown in Figure 1a. All measurements were conducted using the Paios system (Fluxim AG) equipped with a blue monochromatic LED with a peak wavelength at 405 nm. The PST solar cells used in this study were provided by EPFL PV-Lab, Neuchâtel. EQE was determined via transient photocurrent (TPC) measurements at constant applied bias voltage. A sequence of 405 nm LED light pulses with varying intensities (from low to high and reverse) was applied to the devices. The TPC signal of the measured samples is shown in Figure 3a for the case of maximum irradiance (751 W/m^2) and vanishing applied voltage: Each light pulse lasted for 1 s and was preceded by a settling time of 5 s and followed by a recover time of 1 s. The current response was continuously recorded throughout the entire sequence. To minimize external light exposure, the device was covered with a black lid and the entire setup was additionally enclosed with a black towel.

In the raw data, it can be observed that during the light pulse (especially at high intensities) a steady state is not reached within the 1s of illumination. This can be verified in Figure 3b that displays the signal under varying irradiance for a single sample. Consequently, the results might indicate features in the EQE simply because the cells do not reach a quasi-steady state during the measurements. To rule out this possibility, extended measurements with a light pulse duration of 100 s were conducted, and the currents were extracted from different time windows within the pulse. The results show that although a steady state is still not reached, the observed EQE behavior remains consistent (Supporting Information Figure S1a).

The full set of measurements was performed only in the forward direction (that is, from low to high light intensities), as the reverse direction (high to low intensities) yielded nearly identical results in initial tests (see Figure S1b). To ensure homogeneous illumination of the active area, a light mixing rod (LMR) is placed between the LED and the PST solar cell. The LMR leads to a slightly higher increase of the EQE at high light intensities (Figure S1c). To further verify the EQE trend observed at low light intensities, additional measurements were performed using neutral density filters to reduce the intensity from the LED. The two approaches yielded comparable results (Supporting Information Figure S1d). At very low intensities, the filtered data suggest the formation of a plateau, in agreement with the theoretical prediction for the shunt regime as shown below.

4 | Results and Discussion

It is common to assume that the short-circuit current density (J_{SC}) of a tandem solar cell is given by the minimum of both subcell J_{SC}'s due to the exclusively electronic series connection through the recombination junction: Any excess optical generation in one of the two subcells is dissipated through local recombination and hence is lost. This assumption results in a spectral EQE (if no bias light is applied) that is given by the minimum of the subcell absorptances (assuming unity internal quantum efficiency). These are shown in Figure 4a with dashed (perovskite subcell) and dash-dotted (silicon subcell) grey lines, respectively. In the figure, this minimum is highlighted by thick black lines. For a highly luminescent top cell—, i.e., considering increasing values of internal radiative efficiency η_{rad}—however, LC leads to a finite EQE signal at short wavelengths, as the perovskite acts effectively as a down-converter. Dissipation of the excess generation in the top cell through radiative recombination generates photons that are reabsorbed in the Si bottom cell, which results in optical generation and, hence, a secondary photocurrent. In this situation, where an incident photon absorbed in the top cell needs to generate currents in both subcells the EQE is bounded by half of the top-cell EQE (absorptance) in the limit where no emission is lost due to out-coupling at the front. Since the magnitude of the internal emission rate depends exponentially on the top-cell voltage, PR—which increases the local QFLS—boosts the LC signal, as can also be verified in Figure 4a, where the substantially reduced EQE without PR (i.e., without secondary photogeneration due to reabsorption in the perovskite) is indicated with dotted lines. To assess the accuracy of the EC model, full

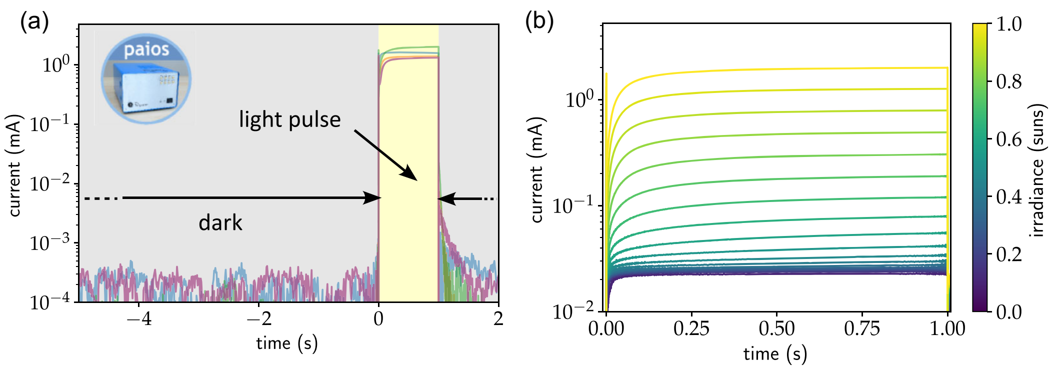

FIGURE 3 | (a) Determination of the monochromatic tandem EQE via transient photocurrent measurements using the all-in-one solar cell characterization platform PAIOS. The relevant signal is extracted from the last 0.2 s under illumination where the current has stabilized. (b) Transient current signal under illumination with varying irradiance.

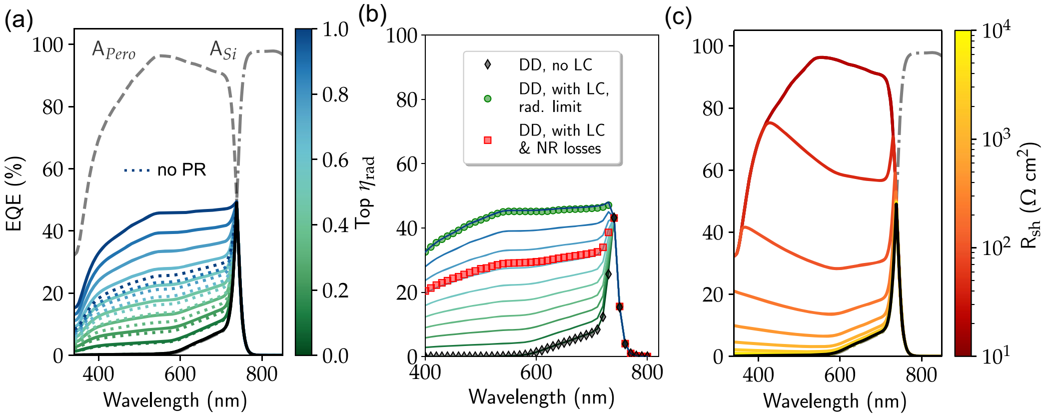

FIGURE 4 | (a) Spectral tandem EQE response under monochromatic illumination as simulated with the EC model as a function of radiative efficiency. Without LC or bottom cell shunts this is given by the minimum of the absorptances, min(A_{pero}, A_{Si}), shown as thick black curves. The LC signal without PR is strongly reduced (dotted lines). (b) The EC model agrees closely with the full DD results for the limit of zero reabsorption (black filled diamonds) and the radiative limit (green filled circles), and nonradiative losses (red filled squares) are well reproduced by a lower radiative efficiency in the top cell. (c) Same as (a), but for the case of a reduced bottom cell shunt resistance.

drift-diffusion (DD) simulations were performed for the situation considered here. As shown in Figure 4b, both the limit of vanishing reabsorption (black diamonds) and the radiative limit (green circles) are perfectly reproduced by the simplified model without transport. Even the DD result for finite LC and nonradiative losses (red squares) can be reproduced remarkably well just by reducing the radiative efficiency in the top cell, which justifies the assumption of unit carrier collection efficiency made in the EC model.

Importantly, a similar signal can result from bottom cell shunt currents, depicted in Figure 4c, as the top cell excess generation does not need to recombine with any bottom-cell charges, and instead flows directly through the ohmic connection. In this case, the top-cell contribution to the bottom-cell EQE is bounded by the top-cell absorptance. However, as evident from their expressions in the EC model, these two contributions to the monochromatic bottom cell EQE exhibit strongly differing subcell voltage dependencies, with the shunt current being proportional to the bottom cell voltage, $J_{\mathrm{sh}} \propto V_{\mathrm{bot}}$, and the LC current proportional to the radiative recombination in the top cell, $J_{\mathrm{LC}} \propto \exp\left[qV_{\mathrm{top}}/(k_{\mathrm{B}}T)\right]$. By employing a variational approach, such as illumination intensity or bias voltage applied to the tandem, the two contributions can hence be distinguished.

Application of the optical model, coupled to the simplified EC model, allows for a prediction of the expected behavior for the PST solar cell device considered, providing a theoretical EQE response as a function of monochromatic (405 nm wavelength) illumination intensity as shown in Figure 5a for the case of the radiative limit. At such short wavelengths it can be assumed that all light is absorbed in the perovskite top cell, as the corresponding bottom-cell absorptance vanishes. The EQE is then evaluated by normalizing the difference of current under illumination and dark current (which vanishes for $V = 0$ V) to the incident photon flux. Selectively turning on LC and shunt conductance via the radiative efficiency and the bottom-cell shunt resistance, respectively, reveals two distinct regimes with respect to the light intensity: At high irradiance levels, LC—which does not depend on light intensity in the absence of nonradiative loss channels—dominates, while shunt current starts contributing only towards low irradiance levels, but eventually dominates the response at the lowest light intensities. Similarly, there is no dependence on external bias voltage for the LC contribution, while such dependence is pronounced for the shunt contribution, especially at low irradiance, as shown in Figure 5b. This can again be understood in terms of the dependence of the photocurrent on subcell voltage: the top-cell voltage governing LC is barely affected in the range of bias voltages applied, whereas the bottom-cell voltage determining the shunt current changes significantly (cf. Supporting Information Figure S2a). To reproduce the measured EQE characteristics—shown are the measurement results for one specific sample as filled triangles—the bottom cell shunt resistance is adjusted (Figure 5c) and the SRH saturation current is increased (Figure 5d), the latter primarily reducing the LC current at lower irradiance. Together, $R_{\mathrm{sh}}^{\mathrm{bot}}$ and $J_{0\mathrm{SRH}}^{\mathrm{top}}$ provide already an excellent fit at low to intermediate light intensity. At high irradiance, however, an overestimation of LC current is found that cannot be removed by increasing bulk SRH recombination without compromising the quality of the fit at lower light intensities. However, if an additional nonradiative loss current with ideality factor close to one is introduced via $J_{0\mathrm{rad}}^{\mathrm{top},*} = J_{0\mathrm{rad}}^{\mathrm{top}}/\mathrm{IQE}^{\mathrm{top}}$ with $\mathrm{IQE}^{\mathrm{top}} < 1$, a perfect fit is possible. Such an LC-quenching loss term can be attributed to interface recombination [37], which was indeed found to represent a limiting process in high-quality perovskite solar cells and which is also included in the full DD model of the tandem.

The resulting fit of the EQE measurements of different samples using the EC model is shown in Figure 6a: although the differences between the samples are sizable—especially at low irradiance—the individual EQE curves are well reproduced, except for a few data points at very low light intensity, where the measurement accuracy is limited. The parameters obtained from the fit are listed in the Supporting Information in Table S1. Next, the EC model was applied to EQE measurements that were conducted at constant V_{bias} values ranging from −0.3 to 1.2 V. The results displayed in Figure 6b confirm that the increase in EQE at high light intensities is independent of V_{bias}, suggesting that shunts in the bottom cell are unlikely to be the cause. At lower light intensities, the EQE varies with V_{bias}, showing a minimum around the V_{OC} of the top cell, which at the lowest intensities is reduced to around 0.85 V

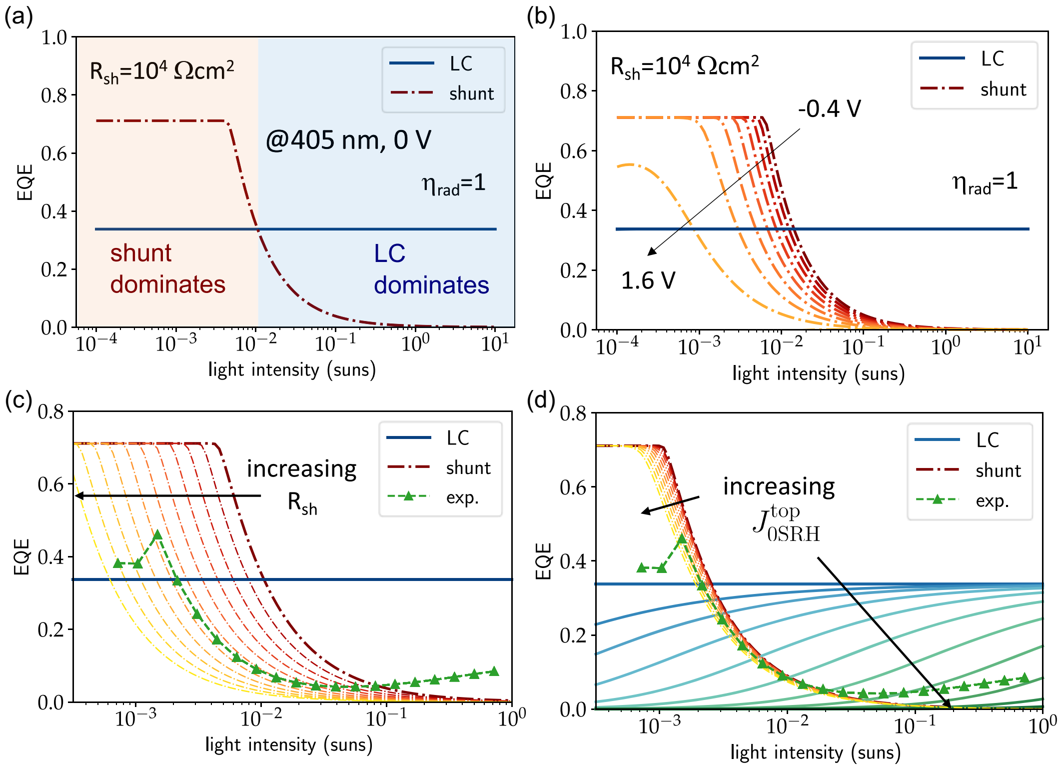

FIGURE 5 | Analysis of the monochromatic EQE signal under illumination with photons of 405 nm wavelength: (a) At the radiative limit, contributions from LC do not depend on intensity and dominate at high irradiances, while contributions from low bottom-cell shunt resistance increase strongly at low irradiance. (b) LC is insensitive to bias voltage, whereas shunt contributions show a strong voltage dependence. (c) Increasing shunt resistance shifts the shunt contribution to lower irradiance levels. (d) Defect-mediated (SRH) recombination reduces LC contributions strongly, especially at lower irradiance.

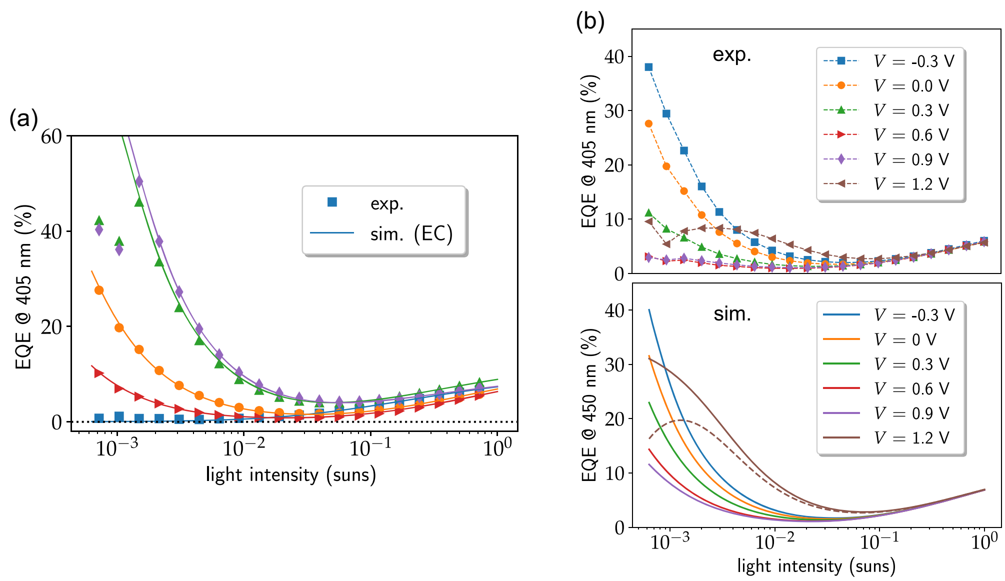

FIGURE 6 | Fitting of the monochromatic EQE experiments with the EC model: (a) Reproduction of the data (filled symbols) from different samples exhibiting a large variation in shunt resistance, but similar LC efficiency. (b) The variation of the shunt contribution to the EQE with bias voltage is reproduced semiquantitatively with the model calibrated at 0 V. The downward bending behavior at maximum voltage can be reproduced with a slight mismatch in the dark current correction (dashed line).

(cf. Supporting Information Figure S2c). Using the EC model for this specific sample calibrated at $V=0$ V, an almost quantitative reproduction of the observed behavior is obtained, which further validates the underlying assumptions regarding the dominant physical processes and of their theoretical description used. Inspecting again the subcell voltages and implied subcell JV curves as a function of applied external voltage reveals that the initial decrease of the shunt contribution is directly related to the

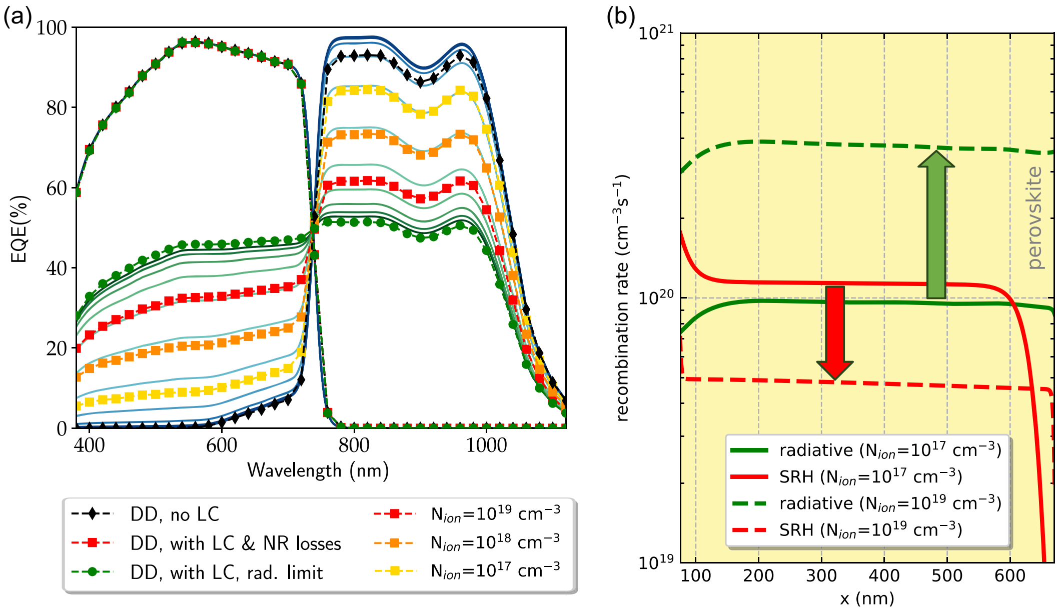

FIGURE 7 | Assessment of LC effects in polychromatic EQE measurements of PSTs with full drift-diffusion simulation (at $V = 0$ V): (a) Subcell EQE computed using broadband bias light tuned to drive the measured cell into current limitation. The bottom cell EQE shows the expected increase at short wavelengths and decrease at long wavelengths. As in the case without bias light, the limits of vanishing reabsorption and exclusively radiative recombination are perfectly reproduced by the EC model (solid lines) in the range of top-cell absorption. However, only the radiative limit agrees fully with DD in the wavelength range of bottom-cell absorption, in the presence of nonradiative losses, these are not fully captured without transport. At reduced concentration of mobile ions, there is a pronounced quenching of the LC signal. (b) Ionic modulation of the radiative and defect-mediated recombination channels explaining the quenching of LC shown in (a).

reduction of the reverse bias on the bottom cell driving the shunt current, while the subsequent increase reflects the evolution of the currents in the dark and under illumination when the bottom cell is driven into the forward bias regime (Supporting Information Figure S2(c) and (d)).

Finally, the model is applied to the more complex case of the polychromatic EQE measurements (i.e., with bias light to selectively measure individual subcell EQE) mentioned in Section 1, but now for a PST. Figure 7a shows the subcell EQE as computed using full drift-diffusion simulation (SETFOS) with AM1.5g bias-light modified with an increment at wavelengths of either 540 nm (for the top cell, blue bias) or 900 nm (for the bottom cell, NIR bias) to enforce current limitation by the measured cell. These two spectra give the reference current under the two bias light conditions. The sensing current is computed with the same spectrum to which a much smaller increment is added at the sensing wavelength that is scanned over the whole range of interest (350–1150 nm). Without loss of generality with respect to the relevant features to be investigated, the simulation is performed for a planar structure. As for the monochromatic EQE, filled black diamonds show the subcell EQEs without reabsorption, while filled circles give the result with PR and LC at the radiative limit. As expected, the top-cell EQE is not affected by reabsorption effects. For the bottom cell, the predictions from the EC model are given with solid lines and the same color code as before for varying radiative efficiency. While at the radiative limit, the models agree also in this case, the presence of nonradiative losses leads to discrepancies at wavelengths corresponding to absorption in the bottom cell. One aspect that is not captured by the EC model, but is considered in the DD simulation, is the impact of mobile ions [38]. In the original calibrated model, a concentration of mobile ionic vacancies of $N_{\text{ion}} = 10^{19}\ \text{cm}^{-3}$ is assumed (red filled squares). If this is reduced to $N_{\text{ion}} = 10^{18}\ \text{cm}^{-3}$ (orange squares) and $N_{\text{ion}} = 10^{17}\ \text{cm}^{-3}$ (yellow squares), the effect of LC is reduced in the entire wavelength range. Since the same change in ionic density does not modify the radiative limit, the effect must be attributed to a modified quenching of LC via defects. Indeed, inspection of the recombination rate profiles inside the perovskite (Figure 7b) reveals that in this particular case, a high concentration of mobile ions leads to strongly reduced SRH recombination due to trap saturation in the bulk and boosts radiative recombination, which provides an intriguing example for the ionic modification of electronic losses [39].

5 | Conclusions

The combined impact of reabsorption effects in state-of-the-art perovskite silicon solar cells is quantified using a combination of monochromatic EQE measurements and rigorous optoelectronic device simulation based on an advanced treatment of internal emission. The approach provides values for LC efficiency and top-to-bottom emission ratio that are consistent with previous experimental findings. Intensity and voltage dependence of the EQE measurements are reproduced quantitatively by an equivalent-circuit model including contributions from reabsorption and finite bottom cell shunt resistance and with effects of bulk and interface recombination. Full drift-diffusion simulations reveal that in this particular case, a high concentration of mobile ions amplifies LC effects as radiative processes are enhanced due to ionic modification of defect-mediated recombination loss.

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Conflicts of Interest

SJZ, DM, SJ, BR, and UA are employed by the company Fluxim AG that commercializes the simulation software tool SETFOS and the measurement setup PAIOS that are used in the study.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] Sometimes, the LC efficiency is defined wrt the *net* emission in the top cell, i.e., after subtraction of the fraction reabsorbed in the top cell (PR). In contrast to Ref. [16], the definition used here does not include the ratio of radiative and nonradiative recombination with ideality factor $n_{\mathrm{id}} = 1$, which renders our LC efficiency a purely optical quantity.

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Supporting Information

Additional supporting information can be found online in the Supporting Information Section. Details on the experimental procedure for the monochromatic EQE measurement, subcell voltages and current-voltage characteristics as obtained from the EC model, EC fitting parameters for the monochromatic EQE measurements. **Supporting Fig. 1:** (a) Independence of the monochromatic EQE from the exact position of the 0.2 second data extraction window within the pulse of 100 s duration (after 10/30/60/90 s). (b) Invariance of the monochromatic EQE with respect to the intensity ramp order (forward: low to high intensity). (c) Impact of the light mixing rod on the monochromatic EQE measurement. (d) Measurements with different neutral density filters shift the EQE to lower irradiance, but do not modify the intensity dependence. **Supporting Fig. 2:** (a) Dependence of the internal subcell voltages on externally applied bias voltage as obtained at the radiative limit of the EC model for the intensity-dependent monochromatic EQE displayed in Fig. 5(b) of the main text. Shown are the values for an intensity of 0.01 suns. While the top-cell voltage governing the LC contribution in the absence of bottom cell shunts (blue solid line) does not vary with applied voltage, the bottom-cell voltage governing the shunt current in the absence of LC (red dash-dotted line) varies linearly with voltage. (b) Subcell voltages for the fit of the experimental EQE curves shown in Fig. 6(b) of the main text. The subcell voltages are shown for the lowest intensity of the monochromatic illumination, where the shunt contributions are strongest. (c) Subcell current-voltage characteristics at lowest irradiance, for the sensing light (solid lines) and in the dark (dashed lines). (d) Excess current generated by the sensing light as a function of externally applied voltage, exhibiting a minimum close to the V_{OC} of the top cell. **Supporting Table 1:** Parameters of the EC model obtained from the fit of the monochromatic EQE measurements for the different perovskite-silicon tandem samples. The IQE^{top} parameter is introduced to reproduce the interface recombination by enhancing the n_{id}=1 recombination current via multiplication of the corresponding saturation current by $1/IQE^{top}$.